Toward Family-centered AI for Supporting Bonding in Caregiving

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Informal caregiving is an important family activity through which family members provide practical assistance, emotional support, and companionship. Meanwhile, caregiving can reshape family relationships by shifting everyday interactions toward tasks, negotiation, and responsibility. Maintaining opportunities for communication, shared identity, intimacy, and mutual understanding is therefore essential for helping caregivers and care recipients remain connected as family members, not only as providers and recipients of care.

This position paper draws on an empirical study investigating how informal caregivers used generative AI-based conversational agents, including ChatGPT, Gemini, and Claude, to support their caregiving. Sixteen informal caregivers participated in the study. Their caregiving relationships included parents caring for children, adult children caring for older parents, siblings, children-in-law caring for parents-in-law, and grandchildren caring for grandparents. Although the study was not originally designed to examine family-centered AI, participants' experiences highlighted AI's potential to help caregivers find constructive ways to communicate with care recipients, reducing conflict and emotional strain. They also underscored the need for family-centered support tailored to family members' capabilities, preferences, roles, and relational dynamics.

Building on these findings, I hope to contribute to the workshop discussion by exploring how family-centered AI might adapt to family dynamics and create opportunities for bonding, while also considering how it could oversimplify relationships, reinforce power imbalances, compromise privacy, or replace meaningful human interaction.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**; **Empirical studies in HCI**.

Additional Key Words and Phrases: Human-centered AI, Informal Caregiving, Qualitative Methods, AI-mediated Communication

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1 Motivation

Families have endured economic, political, religious, and technological change by adapting their forms and functions while remaining an enduring social unit and a "home for the heart" [3]. Within families, caregiving can provide an important context for expressing support, sustaining emotional connection, and reinforcing cohesion [7].

Meanwhile, caregiving can place pressure on the relationships it helps sustain. For example, informal caregivers need to navigate multifaceted responsibilities within and outside caregiving relationships, such as being care recipients' caregivers and loved ones [5, 12], which sometimes result in conflicting expectations and obligations, requiring thoughtful communications with care recipients [5]. Limited guidance for caregiving, communication barriers with care recipients, and uncontrollable situational factors contribute to informal caregivers' emotional strain [1, 2, 8]. In turn, this strain may lead to poor quality of caregivers' relationships with care recipients [4].

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Technology has been proposed as one way to reduce caregiving burdens [11]. Prior work showed that informal caregivers of people living with dementia wanted technology support for routine, burdensome tasks but hoped to keep tasks that fostered personal connections with their care recipients [9]. However, this preferred distinction between instrumental assistance and relational caregiving may become increasingly difficult to negotiate with the emergence of AI kinship, in which AI systems take on relational roles such as caregiver, companion, and grief mediator [6]. As AI capabilities continue to advance, it is important to understand how AI can support family caregiving without displacing valued human interaction and how it can even strengthen relationships between caregivers and care recipients.

2 Relevant Empirical Findings

I hope to contribute to the workshop discussion by drawing on a study I conducted with my advisor. Although the study did not focus on family-centered AI, its findings sparked my interest in how such systems might adapt to family dynamics and support relationships during caregiving. We conducted interviews with sixteen informal family caregivers and, with a think-aloud protocol, asked them to use ChatGPT, Gemini, and Claude to complete a range of caregiving tasks involving logistical work, information seeking, communication with care recipients, and mental health self-care.

Table 1. Demographics of the Participants and Their Care Recipient(s)

ID	Age	Gender	Race	Education (Degree Gained)	Care Recipient(s)	Condition of Care Recipient(s)	Caregiver Role	Caregiving Year(s)
P1	50-65	Female	Other	Graduate or professional degree	Son	Mild cerebral palsy	Prim.	> 20
P2	30-39	Female	Other	Bachelor's degree	Brother	Mild cerebral palsy	Sec.	15-20
P3	30-39	Female	Asian	Bachelor's degree	Son	Children	Prim.	5-10
P4	50-65	Female	Asian	Graduate or professional degree	Mother	Late stage of dementia	Prim.	10-15
P5	40-49	Female	Asian	Bachelor's degree	Daughter, son	Children	Prim.	10-15
P6	40-49	Female	White	Bachelor's degree	Three sons	Teenagers	Prim.	15-20
P7	40-49	Female	Asian	Graduate or professional degree	Parents-in-law, son, daughter	Cancer, depression, diabetes, no chronic condition for children	Sec.	<1
P8	40-49	Male	Asian	Graduate or professional degree	Father-in-law	Early stage of Parkinson's	Sec.	1-3
P9	18-29	Male	Asian	Graduate or professional degree	Father	Diabetes	Sec.	5-10
P10	30-39	Female	Other	Associates or technical degree	Mother	Mental health issue, difficulty moving around, mild cognitive impairment	Sec.	5-10
P11	50-65	Male	White	Graduate or professional degree	Mother	Older adults with short-term injury	Sec.	5-10
P12	30-39	Female	White	Bachelor's degree	Son	Baby	Prim.	1-3
P13	40-49	Male	Asian	Graduate or professional degree	Son	Autism	Prim.	<1
P14	18-29	Female	Asian	High school diploma or GED	Grandmother	Not living, depression	Sec., Sec.	1-3, 5-10
P15	18-29	Female	White	High school diploma or GED	Mother, brother	Stroke, cerebral palsy, spinal muscular atrophy	Prim., Sec., Sec.	1-3, 1-3, <1
P16	18-29	Female	Asian	Bachelor's degree	Father, grandmother	Stroke, diabetes	Sec.	1-3

Notes: Prim. : Primary; Sec. : Secondary

Table 1 shows the participants' demographics and caregiving experiences. Their ages were distributed as 18–29 (25%), 30–39 (25%), 40–49 (31.25%), and 50–65 (18.75%). Twelve participants (75%) were female, and four (25%) were male. Nine participants self-identified as 'Asian' (Korean: 31.25%, Chinese: 12.5%, Indian: 12.5%), four as 'White' (25%), and three as 'Other' races (18.75%). Seven participants (43.75%) held graduate or professional degrees, six (37.5%) held bachelor's degrees, one (6.25%) held an associate or technical degree, and two (12.5%) held a high school diploma or GED. Participants served in primary (50%) and/or secondary (56.25%) caregiving roles. Some participants simultaneously cared for multiple care recipients. Eight participants (50%) cared for parents, six (37.5%) for children, two (12.5%) for siblings, and two (12.5%) for grandparents. Eleven participants (68.75%) cared for loved ones with chronic health conditions affecting physical, metabolic, cognitive, emotional, and social functioning.

2.1 Potential of AI to support bonding in caregiving

Participants identified the potential of AI to support bonding through communication. First, AI could help caregivers navigate interpersonal conflicts by reframing their negative thoughts and encouraging them to consider care recipients' perspectives. For example, AI helped P10 interpret their mother's willingness to go outside with a friend, despite declining P10's invitations, as a positive outcome for care recipients' well-being rather than a personal rejection. Similarly, AI prompted P6 to reconsider sending their son to his room after a conflict, suggesting that separation might make him feel rejected. Second, AI could help caregivers develop more effective communication strategies. Rather than relying on potentially counterproductive approaches, such as threatening, arguing, or avoiding conflicts, P11 and P13 found AI helpful for brainstorming alternative solutions; P7 appreciated its suggestions for communication strategies such as using "I feel" statements instead of "You never help me"; while P2 and P6 found AI helped them craft specific language for difficult conversations "in a way that everybody feels good about". These findings suggest that AI could help caregivers strengthen their relationships with care recipients by fostering understanding of care recipients' perspectives and supporting more empathetic, constructive communication.

2.2 Limitations and Risks of AI for Bonding

However, participants also identified limitations and risks associated with using AI to support communication. AI-generated responses were often perceived as too generic, failing to adequately account for care recipients' cognitive and communicative abilities, communication styles, and personalities. Furthermore, participants emphasized that the appropriateness of communication strategies depended on the existing relationship dynamics between caregivers and care recipients, which AI did not always adequately consider. For example, primary and secondary caregivers sometimes had different expectations regarding their interactions with care recipients, while sons- and daughters-in-law often needed time to build rapport with their parents-in-law. These differences could shape communication dynamics and sometimes contribute to interpersonal tensions. Beyond these limitations, some participants expressed concerns that insensitive AI-generated responses could inadvertently cause harm by exacerbating, rather than resolving, interpersonal conflicts. These findings highlight the need for family-centered AI systems that respond to individual and relational contexts and help caregivers engage in meaningful family interactions.

3 Position and Discussion during Workshop

Family bonds are continually developed through interaction, intimacy, mutual support, shared identity, family stories, rituals, and negotiated responsibilities [10]. I therefore hope to highlight that family-centered AI in caregiving contexts should be evaluated not only in terms of efficiency, accuracy, and burden reduction, but also in terms of relational outcomes. These include whether AI enables meaningful communication, respects the agency of caregivers and care recipients, and preserves or strengthens the bonds between them. This perspective raises important questions about when AI facilitates family interaction, when it risks displacing valued relational work, and how families negotiate the boundaries of AI involvement in care. It further highlights the significance of AI visibility: whether and how family relationships are affected when AI assistance in communication is disclosed, recognizable, or concealed.

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